Markov Chains II

- In this lecture we use R to calculate model probabilities for Markov chains.
- For the former purpose we need to learn about matrices in R, and about matrix multiplication.
- We will then turn to simulating Markov chains using R.
- We will revise writing loops in R and sampling from an arbitrary distribution.
- We will apply Markov chain simulations to model Palmerston North rainfall.

Creating Matricies in R

```
matrix(data = NA, nrow = 1, ncol = 1, byrow = FALSE,
       dimnames = NULL)
```

- Function `matrix` is used to create a matrix.
- First argument, `data`, vector of numbers to populate matrix entries.
- `matrix` requires specification of number of columns and/or rows of resultant matrix.

Creating Matricies in R

```
matrix(data = NA, nrow = 1, ncol = 1, byrow = FALSE,
       dimnames = NULL)
```

`matrix` requires specification of number of columns and/or rows of resultant matrix.

- This is done using `ncol` and `nrow` arguments.
- If just one supplied then other deduced from length of data.
- E.g. if `data` is of length 8 and `nrow=2` then we know matrix must have 4 columns.
- If both `ncol` and `nrow` supplied then elements of `data` recycled if necessary to populate matrix.

Creating Matricies in R

```
matrix(data = NA, nrow = 1, ncol = 1, byrow = FALSE,
       dimnames = NULL)
```

Optional argument `byrow` set to `F` by default.

- Means entries of matrix filled column by column from data vector.
- Set `byrow=T` for entries to be filled row by row.

Example: Fun with Matrices in R

```
matrix(1:4,nrow=2)
```

```
##      [,1] [,2]
## [1,]    1    3
## [2,]    2    4
```

```
matrix(1:4,nrow=2,byrow=T)
```

```
##      [,1] [,2]
## [1,]    1    2
## [2,]    3    4
```

```
matrix(1:4,nrow=4,ncol=4)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    1    1    1    1
## [2,]    2    2    2    2
## [3,]    3    3    3    3
```

Matrix algebra in R

- Addition and subtraction using $+$ and $-$ as expected.
- Standard mathematical matrix multiplication in R specified by $\%*\%$ operator.
  - This standard multiplication is need to compute products like πP for Markov chains.
  - In contrast, recall that $*$ gives element wise multiplication.

Matrix algebra in R

- Also want to compute powers of matrix in standard mathematical way.
  - No function to do so directly in basic version of R.
  - Can achieved by repeated multiplication; e.g. $A\%*\%A\%*\%A$ to find A^3...
  - ... but not a good approach for high powers.
- Use function $\%\hat{}\%$ from package `expm`
  - `expm` is a contributed package that is not part of base R release, so will need installing.
  - Once installed, load with `library(expm)`.

Example More Fun with Matrices in R

```
library(expm)
M <- matrix(1:4,nrow=2)

M^2
```

```
##      [,1] [,2]
## [1,]    1    9
## [2,]    4   16
```

Example More Fun with Matrices in R

```
library(expm)
M <- matrix(1:4,nrow=2)
```

```
M^2
```

```
##      [,1] [,2]
## [1,]    1    9
## [2,]    4   16
```

```
M%*%M
```

```
##      [,1] [,2]
## [1,]    7   15
## [2,]   10   22
```

Example More Fun with Matrices in R

```
M%*%M
```

```
##      [,1] [,2]
## [1,]    7   15
## [2,]   10   22
```

```
M%^%2
```

```
##      [,1] [,2]
## [1,]    7   15
## [2,]   10   22
```

Matrix and Vector Transposition in R

- R assumes by default that vectors are column vectors.
- However, probability distributions for Markov chains usually represented by row vectors.
- Column vector turned into a row vector by transposition.
  - Transposition of a matrix switches rows and columns.
- Function `t` performs transposition of vectors and matrices in R.

Matrix and Vector Transposition in R

```
M <- matrix(c(0, 1, 4, 9),nrow=2)
M
```

```
##      [,1] [,2]
## [1,]    0    4
## [2,]    1    9
```

Function `t` performs transposition of vectors and matrices in R.

```
t(M)
```

```
##      [,1] [,2]
## [1,]    0    1
## [2,]    4    9
```

Example: Calculations for rainfall Markov chain

- Plausible transition matrix for Palmerston North summer rainfall Markov chain is

$$P = \begin{bmatrix} 0.87 & 0.13 \\ 0.50 & 0.50 \end{bmatrix} .$$

- This indicates that $p_{00} = P(X_{t+1} = 0 | X_t = 0) = 0.87$
  - Hence 87% chance dry day followed by another dry day.
  - Wet day equally likely to be followed by a wet or dry day.

Example: Calculations for rainfall Markov chain

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  - Hence 87% chance dry day followed by another dry day.
  - Wet day equally likely to be followed by a wet or dry day.

Questions

1. **Suppose that Monday is dry. What is the probability that Friday is wet?**
2. **Suppose there is a 50% chance that Saturday is wet. What is the probability that the following Saturday will also be wet?**

Example (rainfall MC) continued: Solution to First Question

Questions

1. **Suppose that Monday is dry. What is the probability that Friday is wet?**

- Day $t = 0$ (Monday) is dry, so initial distribution is $\pi^{(0)} = (1, 0)$.
- Want distribution on day $t = 4$ (Friday), given by $\pi^{(4)} = \pi^{(0)} P^4$.
- Do computations in R.

Example (rainfall MC) continued: Solution to First Question

Questions

1. **Suppose that Monday is dry. What is the probability that Friday is wet?**

Start by creating transition matrix.

```
P <- matrix(c(0.87,0.5,0.13,0.5),nrow=2)
P
```

```
##      [,1] [,2]
## [1,] 0.87 0.13
## [2,] 0.50 0.50
```

Example (rainfall MC) continued: Solution to First Question

Want distribution on day $t = 4$ (Friday), given by $\pi^{(4)} = \pi^{(0)} P^4$

```
pi0 <- t(c(1,0))
P4 <- P%^%4
P4
```

```
##           [,1]      [,2]
## [1,] 0.7975181 0.2024819
## [2,] 0.7787765 0.2212235
```

```
pi4 <- pi0%*%P4
pi4
```

```
##           [,1]      [,2]
## [1,] 0.7975181 0.2024819
```

Example (rainfall MC) continued: Solution to First Question

Questions

1. **Suppose that Monday is dry. What is the probability that Friday is wet?**

- Notice the use of $\pi 0 < -t(c(1,0))$ to convert the initial distribution to a row (as opposed to column) vector.
- Package `expm` to access function ^ for matrix powers.
- We conclude that $\pi^{(4)} = (0.798, 0.202)$.
- Hence the probability of a wet Friday is $P(Y_4 = 1) = 0.202$.

Example (rainfall MC) continued: Solution to Second Question

Questions

1. **Suppose there is a 50% chance that Saturday is wet. What is the probability that the following Saturday will also be wet?**

- Set $\pi^{(0)} = (0.5, 0.5)$, since 50% chance day $t = 0$ (Saturday) is wet.
- Want to find probability distribution for day $t = 7$ (following Saturday).

Example (rainfall MC) continued: Solution to Second Question

Questions

1. **Suppose there is a 50% chance that Saturday is wet. What is the probability that the following Saturday will also be wet?**

```
pi0 <- t(c(0.5,0.5))
P7 <- P%^%7
P7
```

```
##              [,1]      [,2]
## [1,] 0.7938467 0.2061533
## [2,] 0.7928974 0.2071026
```

```
pi7 <- pi0%*%P7
pi7
```

Task: Return of the Hopping Frog

- Recall, a frog hops randomly between three lily pads.
- Transition matrix is

$$P = \begin{bmatrix} 0 & \frac{3}{4} & \frac{1}{4} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix}$$

- At time $t = 0$, frog is equally likely to be on any lily pad.
- Use R to compute probability distributions for frog's location at times $t = 2, 4, 10, 20, 100, 1000$.
- What do you notice (if anything)?



For loops in R

- Simulating Markov chains is an iterative process,
- One option is to examine how to write **for** loops.

```
for(i in 1:n){
  instructions
  make_new_var <- i + 3
}
```

- where the instructions are set of commands typically involving `i` in some way.

For loops in R

- Can change counter variable or range. E.g.

```
for(k in c(2,5,7)){
   ...
}
```

- results loop where `k` takes values 2, 5, 7.

Example of a simple for loop in R

We write code to create a loop to calculate vector with first element 100, and where each subsequent element square root of previous one.

```
x <- numeric(10)
x
```

```
##  [1] 0 0 0 0 0 0 0 0 0 0
```

Example of a simple for loop in R

We write code to create a loop to calculate vector with first element 100, and where each subsequent element square root of previous one.

```
x
```

```
##  [1] 0 0 0 0 0 0 0 0 0 0
```

```
x[1] <- 100
for(i in 1:9){
    x[i+1] <- sqrt(x[i])
  }
x
```

```
##  [1] 100.000000  10.000000   3.162278   1.778279   1.333521   1.
##  [7]   1.074608   1.036633   1.018152   1.009035
```

What happened?

- First command creates a numerical vector of length 10, with each element equal to zero.
- We set the first element of `x` to 100.
- As is very common in a loop, the counter `i` is used to index elements of the vector `x`.

```
x[1] <- 100
for(i in 1:9){
    x[i+1] <- sqrt(x[i])
  }
x
```

```
##  [1] 100.000000  10.000000   3.162278   1.778279   1.33352
##  [7]   1.074608   1.036633   1.018152   1.009035
```

Sampling for general distributions

- We have already seen R has functions for simulating from common distributions.
- To simulate Markov chains we will need to sample from non-standard distributions on the state space under consideration.

Sampling for general distributions

- Use command `sample`, which has syntax of the form

```
sample(x, size, replace=F, prob=...)
```

- `x` vector of items from which to sample (here the state space S)
- `size` is number of sampled values that we wish to take;
- `replace` is logical argument indicating whether sampling should be with our without replacement.
- Argument `prob` is optional. If not supplied by the user then the sampling probabilities are uniform. Alternatively, can supply a vector of probabilities corresponding to the elements of `x`.

Example of simple sample

R session below samples from the sequence $\{1, 2, \ldots, 9, 10\}$ in various ways.

```
x <- 1:10
set.seed(1001)
sample(x,size=8,replace=F)
```

```
## [1]  7  3 10  8  9  4  5  1
```

```
sample(x,size=8,replace=T)
```

```
## [1]  6  3  2 10  3  1  8  2
```

```
sample(x,size=3,replace=T,prob=c(0.35,0.25,0.05,0.05,
+ 0.05,0.05,0.05,0.05,0.05,0.05))
```

```
## [1] 10  1  2
```

What happened

- We set the seed of the random number generator so that the results are reproducible.
- Notice that the first sample does not contain any repeats (because elements of `x` are not replaced after selection) while the second sample has three pairs (of 3s, 5s and 9s).
- The final sample places much higher probability on the first two elements of `x`. than the rest. We should therefore not be too surprised that these are then ones sampled.

Sampling Markov Chain Transitions

- Consider Markov chain with state space $S = \{1, 2, \ldots, m\}$ and transition matrix

$$P = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1m} \\ p_{21} & p_{22} & \cdots & p_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ p_{m1} & p_{m2} & \cdots & p_{mm} \end{bmatrix}$$

- Suppose initial distribution is $\pi^{(0)} = (0, 1, 0, \ldots, 0)$ so that the chain starts in state $Y_0 = 2$.
- Transition to time $t = 1$ governed by set of probabilities second row of P.
- Sample single value from this distribution for state at $t = 1$.
- More generally, transitions from i governed by probabilities in ith row of P.

Simulating Palmerston North Summer Rainfall

- Continue to use the Palmerston North summer rainfall Markov chain with transition matrix

$$P = \begin{bmatrix} 0.87 & 0.13 \\ 0.50 & 0.50 \end{bmatrix} .$$

- Will relabel states to be 1 for a dry day and 2 for a rainy day, so state space is of the convenient form $S = \{1, 2\}$.
- Assume that the initial distribution is $\pi^{(0)} = (0.81, 0.19)$.
- Will use R to simulate process for times $t = 0, 1, \ldots, 40$.

Simulating Palmerston North Summer Rainfall

```
set.seed(1001)
y <- numeric(41)
P <- matrix( c( 0.87,0.5,0.13,0.5 ), nrow=2 )
P
```

```
##      [,1] [,2]
## [1,] 0.87 0.13
## [2,] 0.50 0.50
```

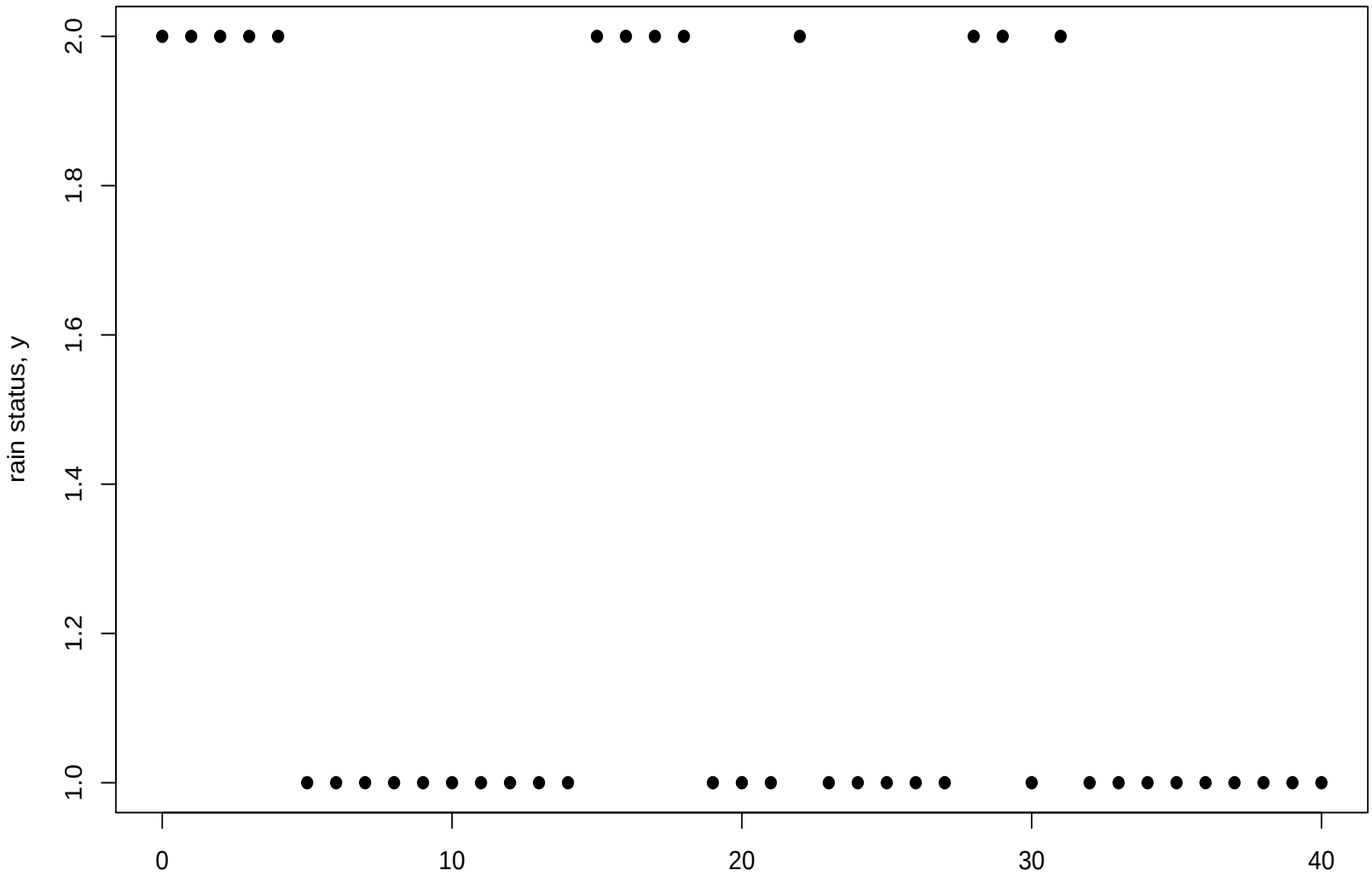
Simulating Palmerston North Summer Rainfall

```
pi0 <- c(0.81,0.19)
y[1] <- sample(1:2,size=1,prob=pi0)
for ( tt in 1:40 ){
  y[tt+1] <- sample(1:2,size=1,prob=P[y[tt],])
}
y
```

```
##  [1] 2 2 2 2 2 1 1 1 1 1 1 1 1 1 1 2 2 2 2 1 1 1 2 1 1 1 1
## [39] 1 1 1
```

Plot it

```
plot(0:40,y,pch=19,xlab="day, t",ylab="rain status, y")
```



How we did it

- Create vector `y` of length 41, to contain the states of the chain for times $t = 0, 1, \ldots, 40$.
  - Note that because vector indexing starts at 1, there is unit offset between the position in `y` and the day t. Classical in the link between computer science and other fields.
  - For example, `y[1]` contains the simulated value of Y_0, and `y[2]` the simulated value of Y_1.
- The random number seed has been set (as usual).
- Initial state of chain, `y[1]`, sampled from initial distribution, the probabilities for which are stored as the vector `pi0`.
- In the loop, the counter `tt` iterates through the days.

Some more points

- Least straightforward part of the code is

```
y[tt+1] <- sample(1:2,size=1,prob=P[y[tt],])
```

- Samples single value from the state space and stores it as the state for the next time point, `y[tt+1]`.
- Sampling probabilities are taken from row number `y[tt]` of the matrix `P`.
- The final command plots simulations. Notice plotting symbol is set to a filled circle by setting the argument `pch=19`.
- Plot looks like a plausible pattern of weather.

Summary

- Matrices in R
- Matrix algebra in R
- Markov chain
- for loops in R
- sampling in R

Task: Back Comes the Hopping Frog

- Recall, a frog hops randomly between three lily pads.
- Transition matrix is

$$P = \begin{bmatrix} 0 & \frac{3}{4} & \frac{1}{4} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix}$$

- At time $t = 0$, frog is equally likely to be on any lily pad.
- Use R to simulate the frog's lily pad hops from time $t = 0$ to $t = 50$ and plot the result.

